رَبِّ اشْرَحْ لِي سَدْرِي وَيَسِّرْ لِي أَمْرِي وَاحْلُلْ عُقْدَةً مِ لن لِسَانِي يَفْقَهُوا قَوْلِي

Rabbish Rahli Sadri wa Yassirli Amri Wahlul Uqadatam Millisani Yafkahu Qawli.

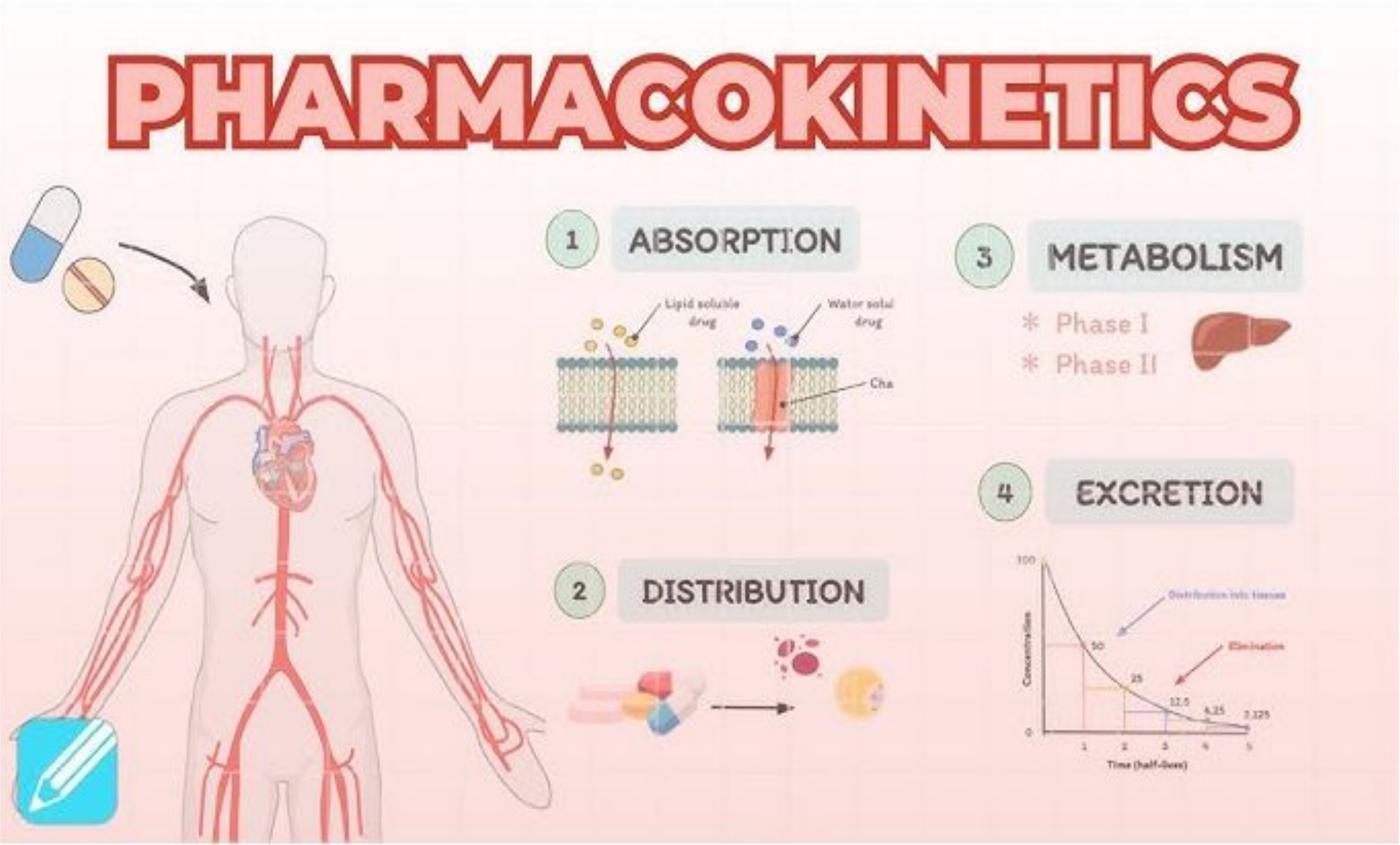
My Lord, Uplift my heart for me, make my task easy, and remove the dullness from my tongue so that they may understand me.
(Surah Ta Ha: 25-26-27-28)

DRUG METABOLISM

DR SANA IRSHAD
PHARMACOLOGY

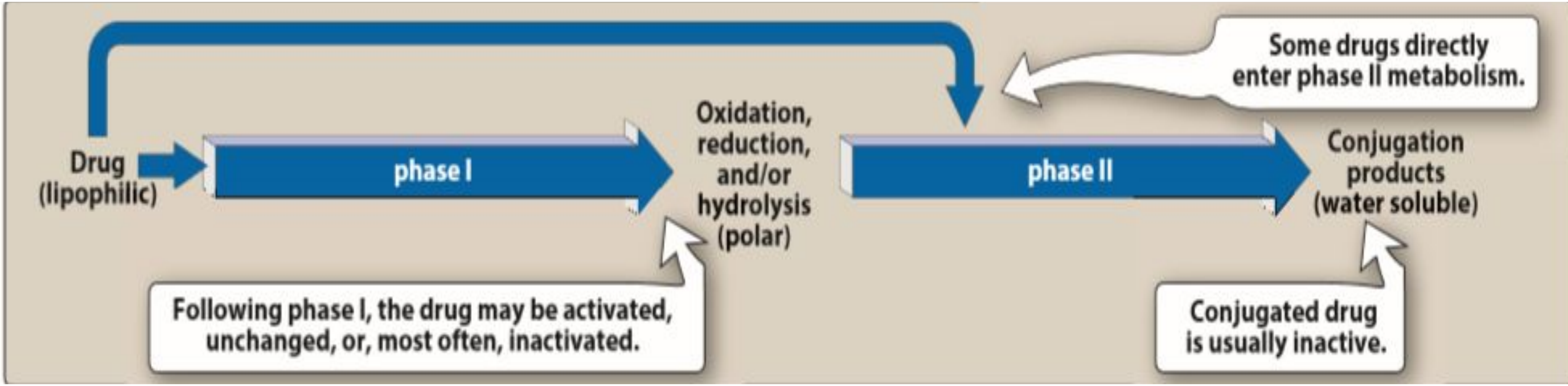
OBJECTIVES

- Define metabolism.
- Discuss the phases of metabolism.
- Discuss cytochrome P 450.
- Discuss the factors affecting metabolism.


PHARMACOKINETICS
1
ABSORPTION
Lipid soluble drug
Water solul drug
Cha
2
DISTRIBUTION
3
METABOLISM
* Phase I
* Phase II
4
EXCRETION
100
50
25
12.5
0
1
2
3
4
5
Time (half-lives)

METABOLISM

- Metabolism leads to production of products with increased polarity, which allows the drug to be eliminated.
- The kidney cannot efficiently eliminate lipophilic. Therefore, lipid-soluble agents are first metabolized into more polar (hydrophilic) substances in the liver via two general sets of reactions, called phase I and phase II.


Some drugs directly enter phase II metabolism.
Drug (lipophilic)
phase I
Oxidation, reduction, and/or hydrolysis (polar)
phase II
Conjugation products (water soluble)
Following phase I, the drug may be activated, unchanged, or, most often, inactivated.
Conjugated drug is usually inactive.

PHASE 1 REACTION

- Convert lipophilic drugs into more polar molecules by introducing or unmasking a polar functional group, such as –OH or –NH2.
- Phase I reactions usually involve reduction, oxidation, or hydrolysis.

- Phase I reactions utilizing the P450 system:

The phase I reactions most involved in drug metabolism are catalyzed by the cytochrome P450 system (also called microsomal mixed-function oxidases).

These enzymes are found in high concentrations in the smooth endoplasmic reticulum of the liver.

It is important for the metabolism of many endogenous compounds (such as steroids, lipids) and for the biotransformation of exogenous substances (xenobiotics).

CYTOCHROME P 450

Cytochrome P450, designated as CYP, is a superfamily that are located in most cells, but primarily in the liver and GI tract.

Nomenclature: The family name is indicated by the Arabic number that follows CYP, and the capital letter designates the subfamily, for example, CYP3A. A second number indicates the specific isozyme, as in CYP3A4.

Four isozymes are responsible for the vast majority of P450-catalyzed reactions. They are CYP3A4/5, CYP2D6, CYP2C8/9, and CYP1A2.

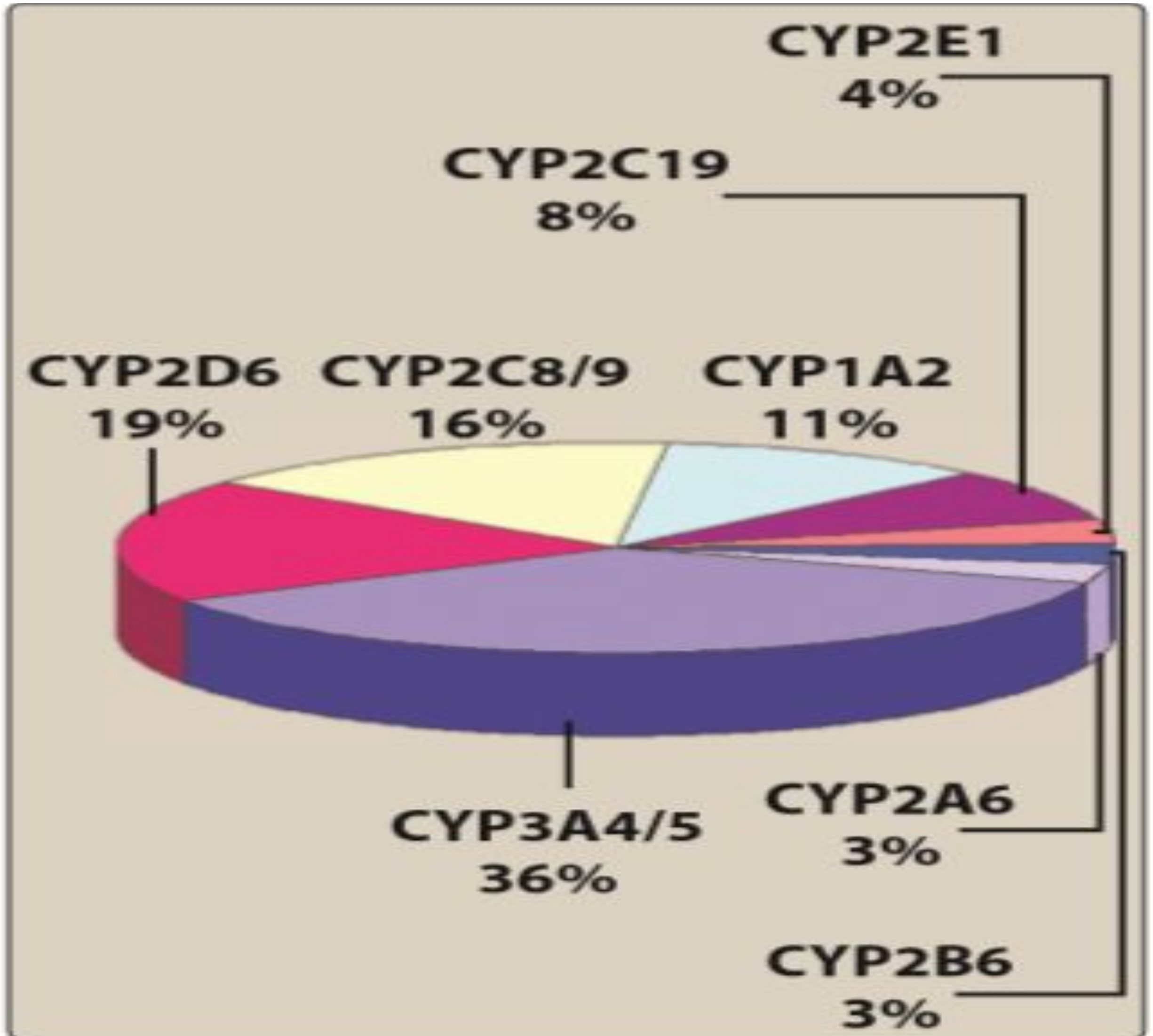


TABLE 4–1 **Examples of phase I drug-metabolizing reactions.**

Reaction Type	Typical Drug Substrates
Oxidations, P450 dependent	
Hydroxylation	Amphetamines, barbiturates, phenytoin, warfarin
N-dealkylation	Caffeine, morphine, theophylline
O-dealkylation	Codeine
N-oxidation	Acetaminophen, nicotine
S-oxidation	Chlorpromazine, cimetidine, thioridazine
Deamination	Amphetamine, diazepam
Oxidations, P450 independent	
Amine oxidation	Epinephrine
Dehydrogenation	Chloral hydrate, ethanol
Reductions	Chloramphenicol, clonazepam, dantrolene, naloxone
Hydrolyses	
Esters	Aspirin, clofibrate, procaine, succinylcholine
Amides	Indomethacin, lidocaine, procainamide

- Phase I reactions not involving the P450 system:

These include amine oxidation (for example, oxidation of catecholamines or histamine), alcohol dehydrogenation (for example, ethanol oxidation), esterases (for example, metabolism of aspirin in the liver), and hydrolysis (for example, of procaine).

PHASE 2 REACTION

- Phase II reactions are synthetic reactions that involve addition (conjugation) of subgroups to —OH, —NH_2, and —SH functions on the drug molecule.
- The subgroups that are added include glucuronate, acetate, glutathione, glycine, sulfate, and methyl groups.
- Most of these groups are relatively polar and make the product less lipid-soluble than the original drug molecule.

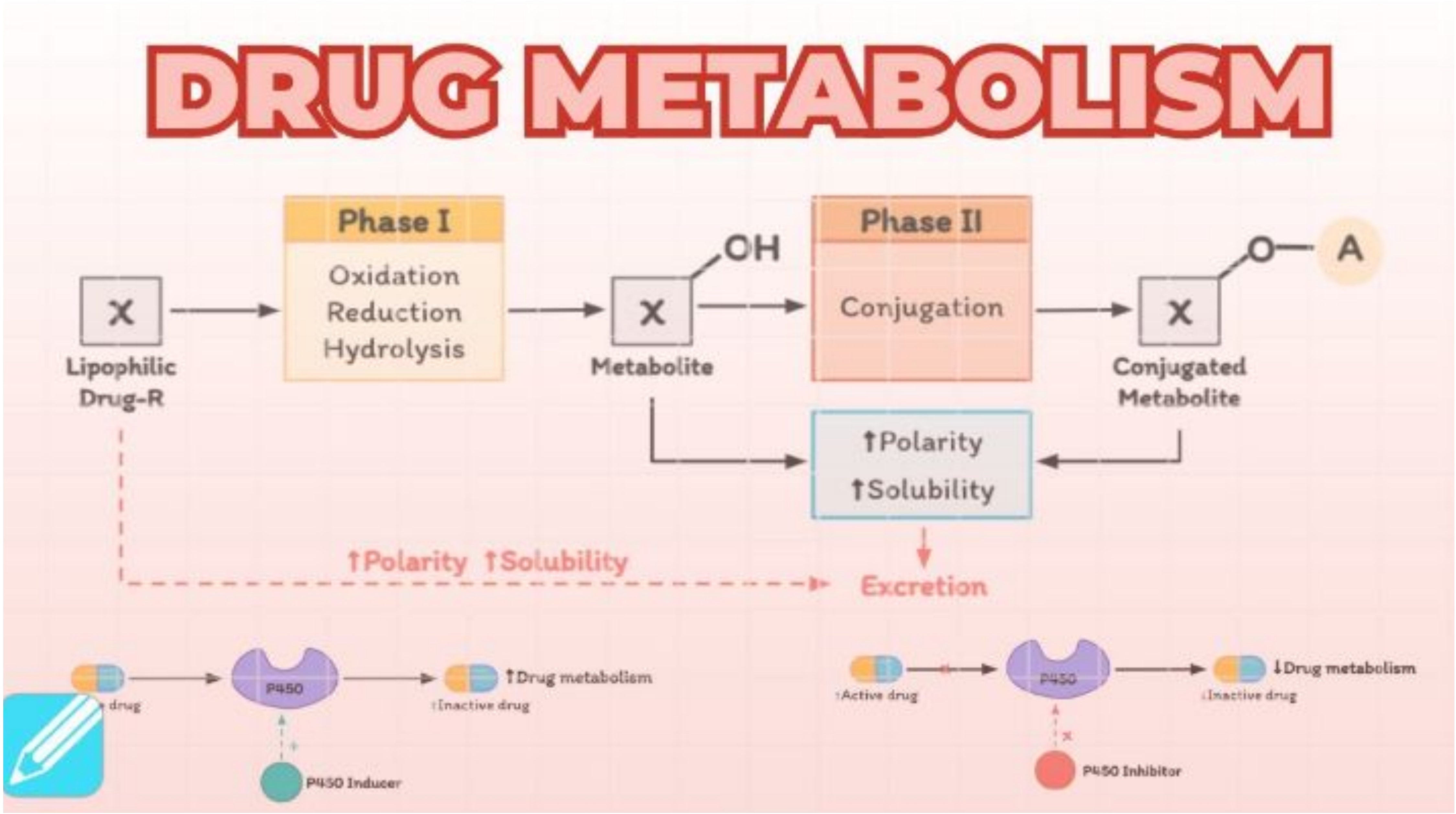

DRUG METABOLISM
Phase I
Oxidation
Reduction
Hydrolysis
Phase II
Conjugation
X
Lipophilic Drug-R
X
OH
Metabolite
X
O
A
Conjugated Metabolite
↑Polarity
↑Solubility
↑Polarity ↑Solubility
Excretion
P450
↑Drug metabolism
↑Inactive drug
P450 Inducer
↑Active drug
P450
↓Drug metabolism
↓Inactive drug
P450 Inhibitor

TABLE 4–2 **Examples of phase II drug-metabolizing reactions.**

Reaction Type	Typical Drug Substrates
Glucuronidation	Acetaminophen, diazepam, digoxin, morphine, sulfamethiazole
Acetylation	Clonazepam, dapsone, isoniazid, mescaline, sulfonamides
Glutathione conjugation	Ethacrynic acid, reactive phase I metabolite of acetaminophen
Glycine conjugation	Deoxycholic acid, nicotinic acid (niacin), salicylic acid
Sulfation	Acetaminophen, methyldopa
Methylation	Dopamine, epinephrine, histamine, norepinephrine, thiouracil

Adapted, with permission, from Katzung BG, editor: *Basic & Clinical Pharmacology*, 12th ed. McGraw-Hill, 2012.

DETERMINANTS OF BIOTRANSFORMATION RATE

A. Genetic Factors: P450 enzymes exhibit considerable genetic variability among individuals and racial groups. Variations in P450 activity may alter drug efficacy and the risk of adverse events. CYP2D6, in particular, has been shown to exhibit genetic polymorphism. . CYP2D6 mutations result in very low capacities to metabolize substrates. Some individuals, for example, obtain no benefit from the codeine, because they lack the CYP2D6 enzyme that activates the drug.

B. Effects of Other Drugs: Coadministration of certain agents may alter the disposition of many drugs. Mechanisms include the following: 1. Enzyme induction—Induction (increased rate and extent of metabolism) usually results from increased synthesis of

C. Enzyme induction

- Induction (increased rate and extent of metabolism) usually results from increased synthesis of cytochrome P450 drug-oxidizing enzymes in the liver as well as the cofactor, heme.
- Drugs and other xenobiotics that increase enzyme activity are known as inducers.
- The most common strong inducers of drug metabolism are carbamazepine, phenobarbital, phenytoin, and rifampin.
- **For example**, rifampin, an antituberculosis drug, significantly decreases the plasma concentrations of human immunodeficiency virus (HIV) protease inhibitors, thereby diminishing their ability to suppress HIV replication.

TABLE 4–3 A partial list of drugs that significantly induce P450-mediated drug metabolism in humans.

CYP Family Induced	Important Inducers	Drugs Whose Metabolism Is Induced
1A2	Benzo[*a*]pyrene (from tobacco smoke), carbamazepine, phenobarbital, rifampin, omeprazole	Acetaminophen, clozapine, haloperidol, theophylline, tricyclic antidepressants, (*R*)-warfarin
2C9	Barbiturates, especially phenobarbital, phenytoin, primidone, rifampin	Barbiturates, celecoxib, chloramphenicol, doxorubicin, ibuprofen, phenytoin, chlorpromazine, steroids, tolbutamide, (*S*)-warfarin
2C19	Carbamazepine, phenobarbital, phenytoin, rifampin	Diazepam, phenytoin, topiramate, tricyclic antidepressants, (*R*)-warfarin
2E1	Ethanol, isoniazid	Acetaminophen, enflurane, ethanol (minor), halothane
3A4	Barbiturates, carbamazepine, corticosteroids, efavirenz, phenytoin, rifampin, pioglitazone, St. John's wort	Antiarrhythmics, antidepressants, azole antifungals, benzodiazepines, calcium channel blockers, cyclosporine, delavirdine, doxorubicin, efavirenz, erythromycin, estrogens, HIV protease inhibitors, nefazodone, paclitaxel, proton pump inhibitors, HMG-CoA reductase inhibitors, rifabutin, rifampin, sildenafil, SSRIs, tamoxifen, trazodone, vinca alkaloids

SSRIs, selective serotonin reuptake inhibitors.

D. Enzyme inhibition

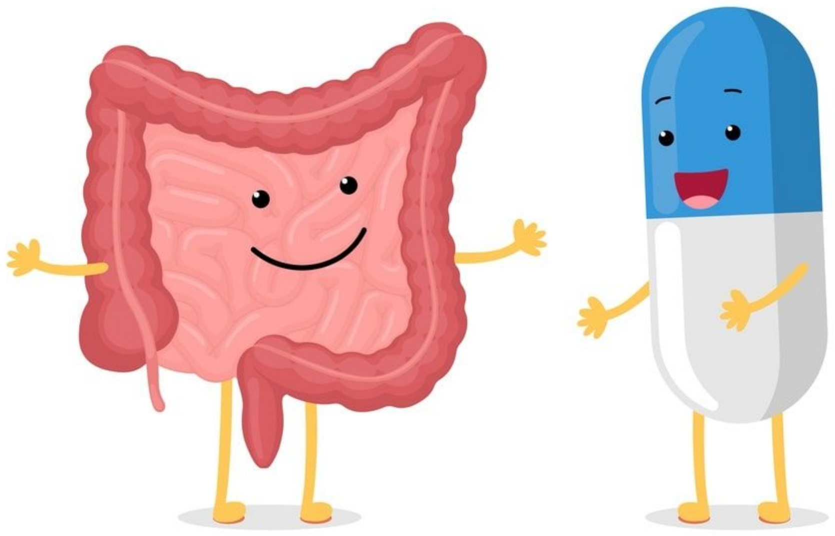
- Inhibition of CYP isozyme activity is an important source of drug interactions that lead to serious adverse events. For example amiodarone, cimetidine present in grapefruit juice, azole antifungals, and the HIV protease inhibitor ritonavir.
- Metabolism may also be decreased by pharmacodynamic factors such as a reduction in blood flow to the metabolizing organ (eg, propranolol reduces hepatic blood flow).
- For example, omeprazole is a potent inhibitor of three of the CYP isozymes responsible for warfarin metabolism. If the two drugs are taken together, plasma concentrations of warfarin increase, which leads to greater anticoagulant effect and increased risk of bleeding.

TABLE 4–4 A partial list of drugs that significantly inhibit P450-mediated drug metabolism in humans.

CYP Family Inhibited	Inhibitors	Drugs Whose Metabolism Is Inhibited
1A2	Cimetidine, fluoroquinolones, grapefruit juice, macrolides, isoniazid, zileuton	Acetaminophen, clozapine, haloperidol, theophylline, tricyclic antidepressants, (*R*)-warfarin
2C9	Amiodarone, chloramphenicol, cimetidine, isoniazid, metronidazole, SSRIs, zafirlukast	Barbiturates, celecoxib, chloramphenicol, doxorubicin, ibuprofen, phenytoin, chlorpromazine, steroids, tolbutamide, (*S*)-warfarin
2C19	Fluconazole, omeprazole, SSRIs	Diazepam, phenytoin, topiramate, (*R*)-warfarin
2D6	Amiodarone, cimetidine, quinidine, SSRIs	Antiarrhythmics, antidepressants, beta blockers, clozapine, flecainide, lidocaine, mexiletine, opioids
3A4	Amiodarone, azole antifungals, cimetidine, clarithromycin, cyclosporine, diltiazem, erythromycin, fluoroquinolones, grapefruit juice, HIV protease inhibitors, metronidazole, quinine, SSRIs, tacrolimus	Antiarrhythmics, antidepressants, azole antifungals, benzodiazepines, calcium channel blockers, cyclosporine, delavirdine, doxorubicin, efavirenz, erythromycin, estrogens, HIV protease inhibitors, nefazodone, paclitaxel, proton pump inhibitors, HMG-CoA reductase inhibitors, rifabutin, rifampin, sildenafil, SSRIs, tamoxifen, trazodone, vinca alkaloids

SSRIs, selective serotonin reuptake inhibitors.

EXCRETION AND FACTORS AFEECTING EXCRETION AND ELIMINATION



OBJECTIVES

- Define clearance and half life.
- Discuss the kinetics of elimination.
- Define steady state concentration.

CLEARANCE

The ratio of the rate of elimination of a drug to the concentration of the drug in the plasma or blood

$$CL = \frac{\text{Rate of elimination of drug}}{\text{Plasma drug concentration}}$$

- Units = volume/time
- For a drug eliminated with first order kinetics clearance is constant
- For a drug eliminated with zero order kinetics clearance is not constant
- Clearance depends on blood flow and the conditions of the organs of elimination in the patient

HALF LIFE ($t_{1/2}$)

The time required for the amount of drug in the body or blood to fall by 50%

$$t_{1/2} = \frac{0.693 \times Vd}{CL}$$

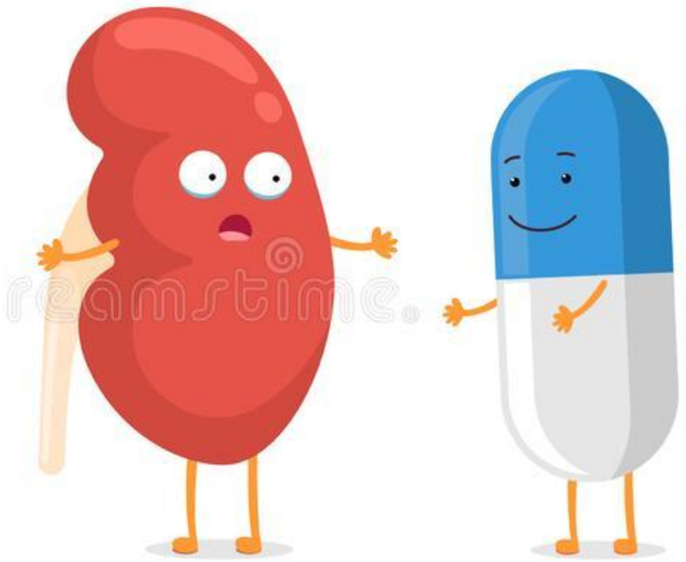
(units = time)

Factors increasing drug half life

- Decreased renal plasma flow
- Decreased extraction ratio
- Decreased metabolism

EXCRETION

Excretion is defined as the process by which drug or metabolites are irreversibly transferred from the internal to the external environment through renal or non renal routes.



TYPES OF EXCRETION

- **RENAL**

- **NON RENAL**

Biliary excretion

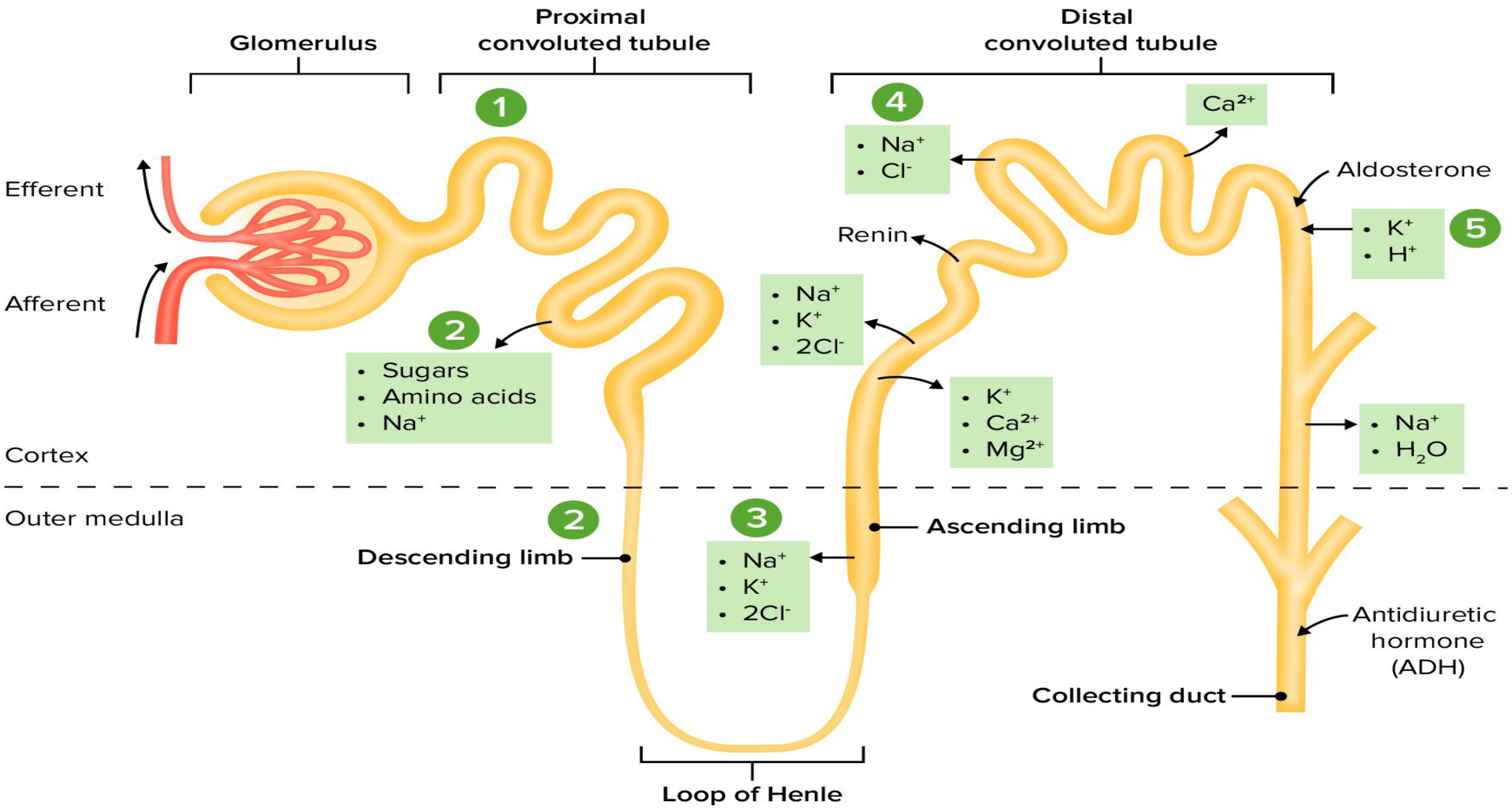
Pulmonary excretion

Salivary excretion

Mammary excretion

Skin excretion

Renal Elimination Of Drug



Renal elimination of a drug

Elimination of drugs via the kidneys into urine involves the processes of glomerular filtration, active tubular secretion, and passive tubular reabsorption.

1. Glomerular filtration

- Drugs enter the kidney through renal arteries which divide to form a glomerular capillary plexus
- Free drug (not bound to albumin) flows through the capillary slits
- GFR is normally about 125 mL/min (may diminish in kidney disease)
- Variations in GFR and protein binding of drugs do affect this process

2. Proximal tubular secretion

- Drugs that were not transferred to the glomerular filtrate leave the glomeruli through efferent arterioles, which divide to form a capillary plexus surrounding the nephric lumen in the proximal tubule.
- Secretion occurs in the proximal tubules by two energy-requiring active transport systems:
- for *anions* (for example, deprotonated forms of weak acids)
- for *cations* (for example, protonated forms of weak bases).

3. **Distal tubular reabsorption**

- As a drug moves toward the distal convoluted tubule, its concentration increases
- The drug, if uncharged, may diffuse out of the nephric lumen, back into the systemic circulation
- Manipulating the urine pH to increase the fraction of ionized drug in the lumen may be done to minimize the amount of back diffusion and increase the clearance of an undesirable drug.
- Weak acids can be eliminated by **alkalinization** of the urine, whereas elimination of weak bases may be increased by **acidification** of the urine. This process is called **ion trapping**

For example

A patient presenting with phenobarbital (weak acid) overdose

↓

can be given bicarbonate

↓

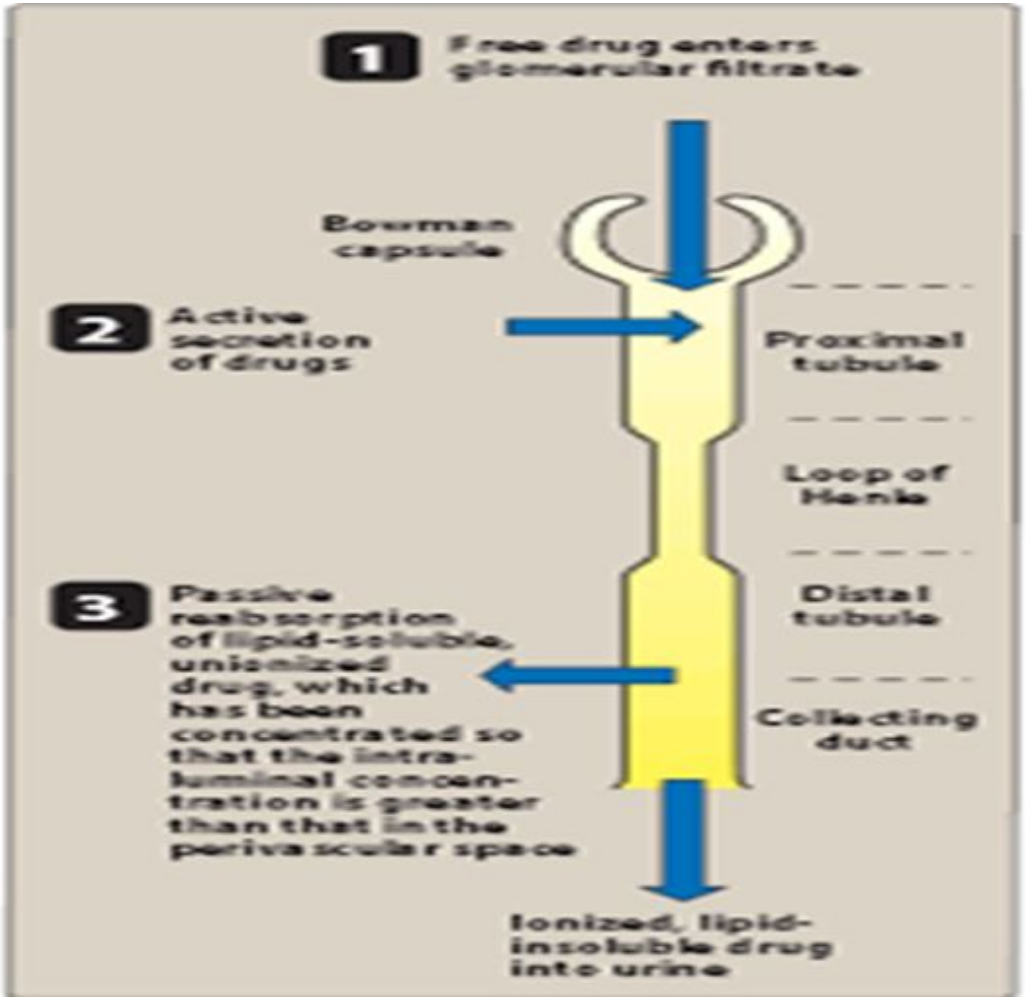
which alkalinizes the urine

↓

keeps the drug ionized

↓

thereby decreasing its reabsorption


1 Free drug enters glomerular filtrate
Bowman capsule
2 Active secretion of drugs
Proximal tubule
Loop of Henle
Distal tubule
3 Passive reabsorption of lipid-soluble, unionized drug, which has been concentrated so that the intraluminal concentration is greater than that in the perivascular space
Collecting duct
Ionized, lipid-insoluble drug into urine

FACTORS AFFECTING EXCRETION OF DRUGS

- Physiochemical properties of drugs (mol size, lipid solubility)
- Plasma concentration of drug (filtered and reabsorbed , secreted)
- Distribution and binding characteristics of drug
- Blood flow to the kidney
- Biological factor (old age altered GFR and tubular function)
- Drug interaction (alteration in protein binding, pH)
- Disease state

ELIMINATION

- Drugs are eliminated by reducing their concentration or amount in the body. This occurs when the drug is inactivated by metabolism or excreted from the body. Conversion to an inactive metabolite is a form of elimination.
- A drug may be eliminated by metabolism long before the modified molecule as excreted from the body

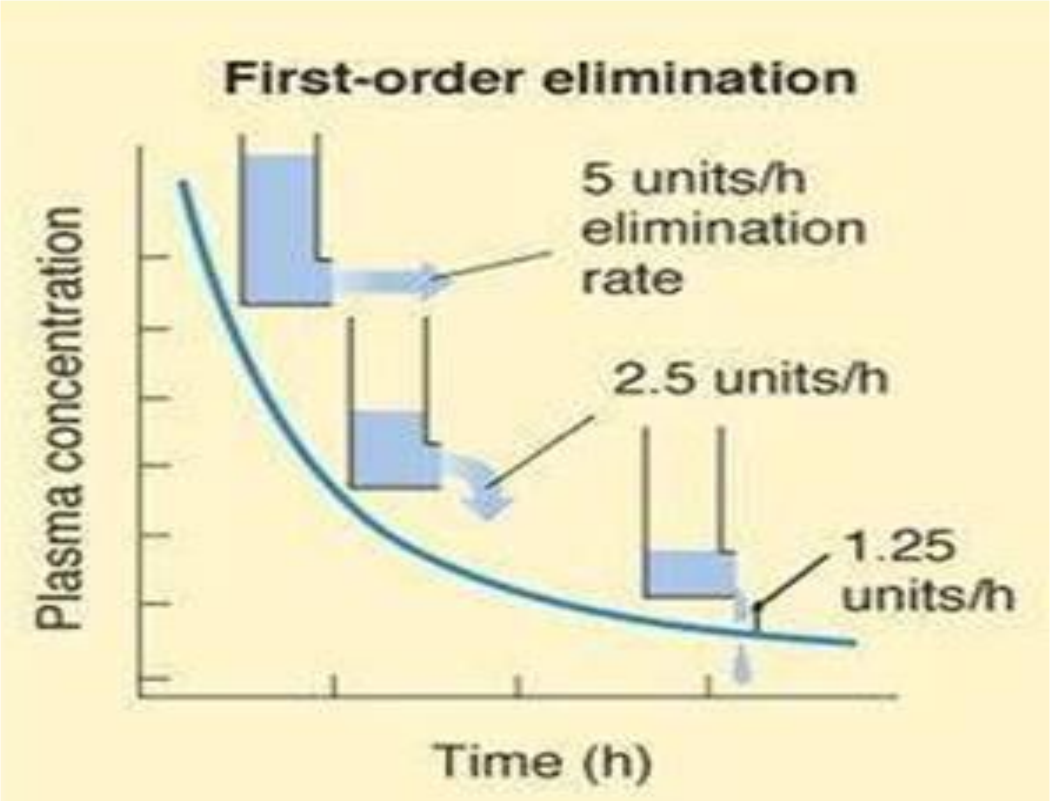
- Action of many drugs terminated before they are excreted because they are metabolize to inactive derivatives
- Prodrugs are inactive as administered and must be metabolized in the body to become active
- Many drugs are active as administered and have active metabolites as well
- Some drugs are not modified by body they continue to act until they are excreted
- Small number of drugs continue irreversibly with their receptors so that disappearance from the body is not equivalent to cessation of drug action

ELIMINATION KINETICS

- The rate of elimination of drugs may be zero order or first order

FIRST ORDER ELIMINATION

- Rare of elimination is proportional to the concentration
- Drug concentration in plasma decreases with time
- Half life is constant regardless of the amount of drug in the body
- The concentration of such drug in the blood will decrease by 50% for every half life
- Most drugs demonstrates first order kinetics

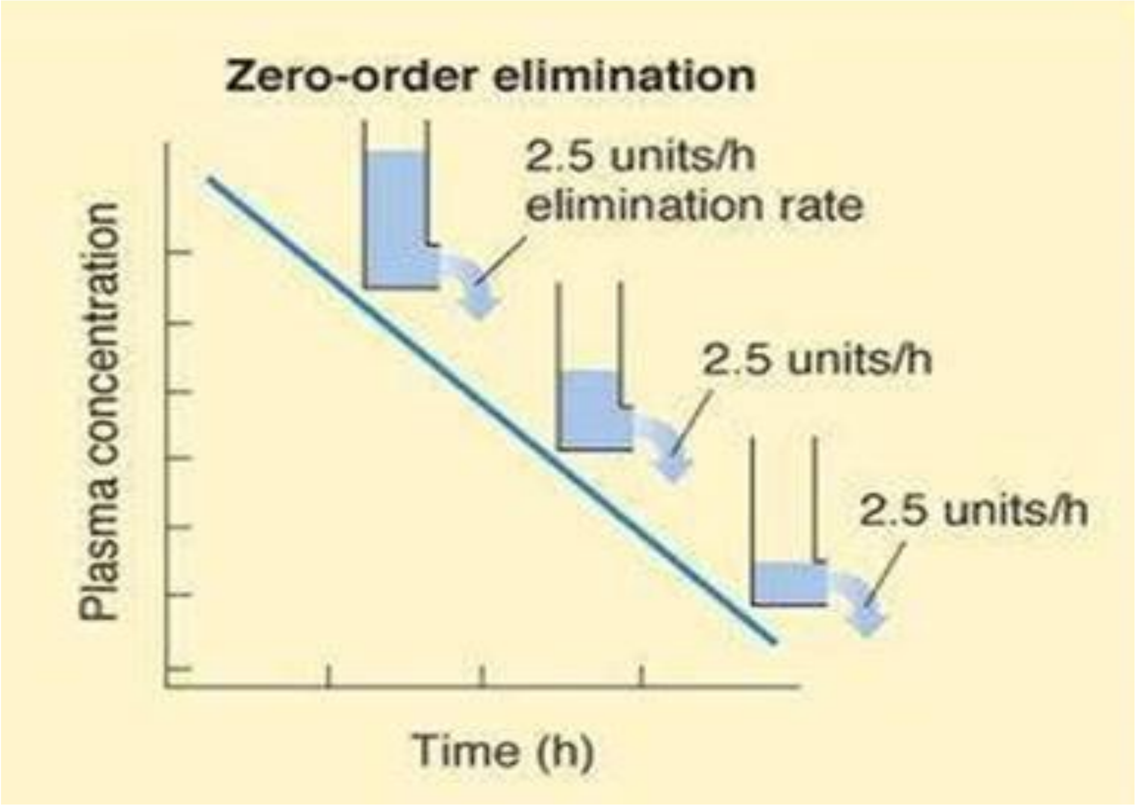


- 80mg $\xrightarrow{4h}$ 40mg $\xrightarrow{4h}$ 20mg $\xrightarrow{4h}$ 10mg $\xrightarrow{4h}$ 5mg

ZERO ORDER ELIMINATION

- Rate of elimination is constant regardless of concentration
- Concentration of these drugs in plasma decreases in a linear fashion over time
- Do not have constant half life
- For example: ethanol, phenytoin and aspirin at high therapeutic or toxic doses.

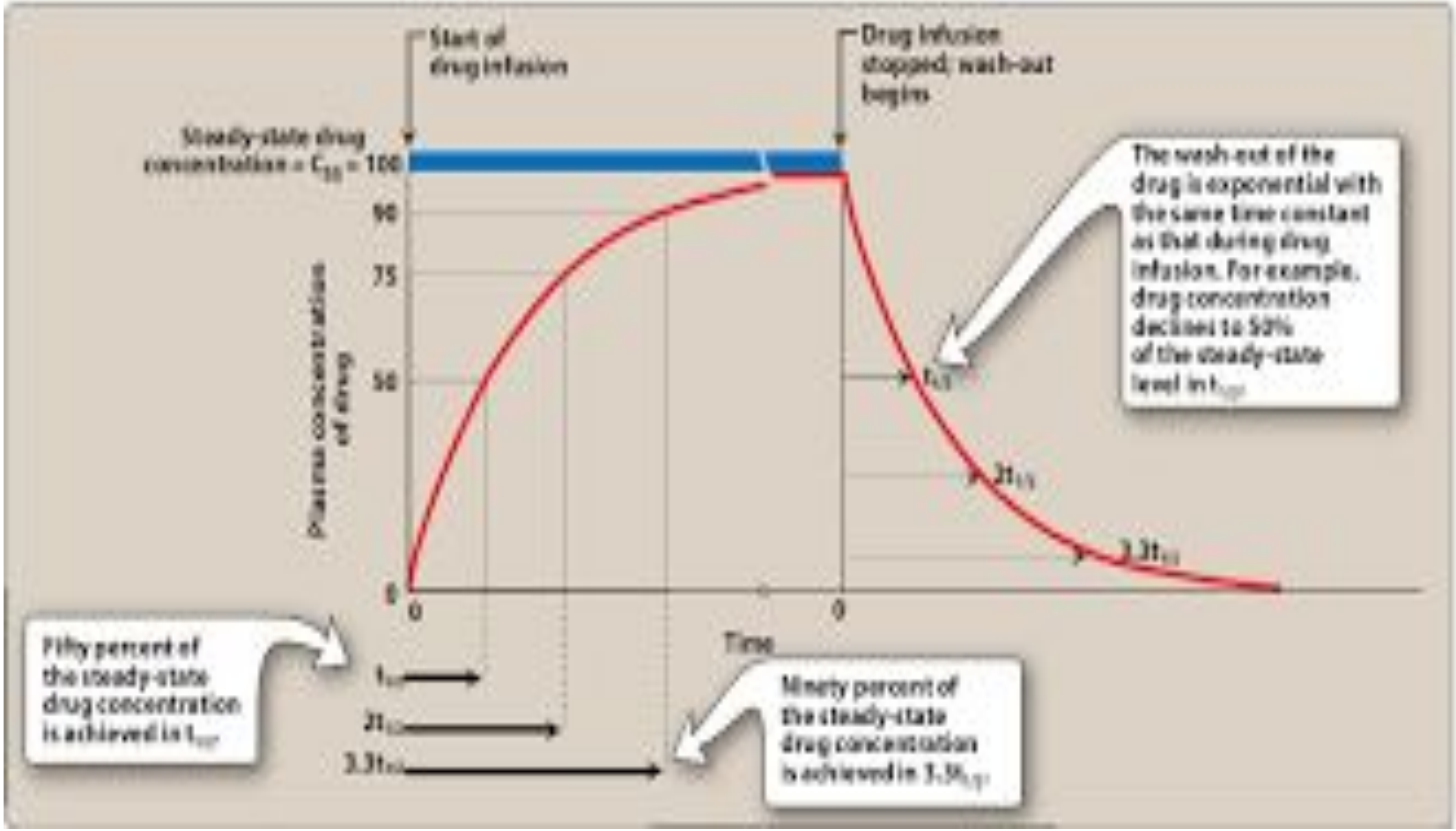
- 80mg $\xrightarrow[4h]{}$ 70mg $\xrightarrow[4h]{}$ 60mg $\xrightarrow[4h]{}$ 50mg $\xrightarrow[4h]{}$ 40mg



STEADY STATE CONCENTRATION

- The condition in which the rate of drug elimination equals the rate of administration.
- The effect of a drug at 87-90% of its steady state concentration is clinically indistinguishable from the steady state effect thus 3-4 half lives of dosing at a constant rate are considered adequate to produce the effect to be expected at steady state.

STEADY STATE CONCENTRATION



THANKS………..


Sir, This way please
(out of the body)
We are not Real Brothers
We are Renal Brothers